

Fergus Donnan

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Research Interests

Galaxy Evolution, Dust Obscuration/Evolution, SMBH Growth, AGN/Star-formation, PAHs, Molecules, AGN Outflows, Accretion Disks

Education

- 2021–2025 **DPhil (PhD) Astrophysics**, *University of Oxford*, Oxford, UK
"The Obscured Universe: Tracing Star Formation and AGN activity across cosmic time."
Supervisors: Dimitra Rigopoulou, Ismael García-Bernete.
- 2016–2021 **MPhys (Hons) Astrophysics**, *University of St Andrews*, St Andrews, UK, 1st class
"Echo Mapping of AGN Accretion."
Supervisors: Keith Horne, Juan Hernandez Santisteban.

Research Positions

- 2025 - present **Postdoc**, *University of California San Diego*, San Diego, USA
- 2025 **Postdoc**, *University of Oxford*, Oxford, UK
- 2023 **Visiting Graduate Student**, *CAB ESAC*, Madrid, Spain
- 2020 **LEAPS Research Student**, *Leiden Observatory*, Leiden, Netherlands
- 2019 **Summer Research Project**, *University of St Andrews*, St Andrews, UK

Awards & Scholarships

- 2021 **Principal's Scholarship for Academic Excellence**, Awarded annually to the fifty final-year students, at the University of St Andrews, whose academic performances are the highest in their respective faculties.
- 2020 **Scott Lang Prize**, Most outstanding student in the Senior Honours class in Astronomy and Astrophysics.
- 2016–2021 **Deans List**, For averaging a 1st class across the academic year.
- 2020 **Medal (Astrophysics Fourth level)**
- 2019 **Medal (Astrophysics Third level)**
- 2019 **Carnegie Trust Vacation Scholarship**, Awarded to fund a place on an undergraduate research internship programme.
- 2018 **Medal (Astrophysics Second level)**

Presentations

- 2026 **Exploring the Aromatic Universe in the JWST era**, London, Canada, Contributed Talk
Spatially resolved PAH spectroscopy in the era of JWST: From kinematics of PAHs to resolved maps at cosmic noon.
- 2026 **Oxford Galaxy Evolution Seminar**, Oxford, UK, Invited Talk
Organic dust grains across cosmic scales: From resolved PAH maps at cosmic noon to velocity maps of PAHs in AGN outflows
- 2026 **Charting Cosmic Dawn**, Copenhagen, Denmark, Contributed Talk
PAHSPECS: Spatially Resolved PAH Spectroscopy at cosmic noon with MIRI MRS
- 2025 **IAC Galaxy Coffee**, Tenerife, Spain, Invited Talk
Spatially resolved PAH spectroscopy in the era of JWST: From kinematics of PAHs to resolved maps at cosmic noon.
- 2025 **CAB Extragalactic Meeting**, Madrid, Spain, Invited Talk
The Dust Obscured Universe: Unveiling Star-formation and AGN activity across cosmic time with mid-infrared spectroscopy.
- 2025 **PanDust 2025**, Tucson, USA, Contributed Poster
Spatially resolved PAH Spectroscopy in the era of JWST: From kinematics of PAHs to resolved maps at cosmic noon.
- 2025 **UCSD Astrophysics Colloquium**, San Diego, USA, Invited Talk
The Dust Obscured Universe: Unveiling Star-formation and AGN activity across cosmic time with mid-infrared spectroscopy.
- 2025 **Oxford Galaxy Evolution Seminar**, Oxford, UK, Invited Talk
The Obscured Growth of SMBHs: Revealing the nature of the most obscured galaxy nuclei in the mid-infrared.
- 2025 **Dusting Off the Secrets of the Cosmos with PRIMA Space IR Telescope**, Marseille, France, Contributed Talk
A Census of the Most Obscured Galaxy Nuclei over Cosmic Time to be revealed by PRIMA.
- 2025 **Behind a curtain of dust, V**, Kiruna, Sweden, Contributed Talk
Peeling Back the Layers of Extinction of the Most Obscured Nuclei in the Era of JWST: What is their nature and evolution?
- 2024 **Cyprus Astrophysics Workshop 2024: Luminous Infrared galaxies in the JWST era**, Nicosia, Cyprus, Invited Talk
Peeling Back the Layers of Extinction of the Most Obscured Galaxy Nuclei in the Era of JWST.
- 2024 **Oxford Astrophysics Colloquium**, Oxford, UK, Early Career Showcase
The Obscured Growth of SMBHs: Peeling Back the Layers of Extinction of the Most Obscured Galaxy Nuclei in the Era of JWST.
- 2024 **Cosmic Odysseys**, Crete, Greece, Invited Talk
Peeling back the layers of extinction of dusty galaxies in the era of JWST.
- 2023 **European Astronomical Society (EAS)**, Krakow, Poland, Contributed Talk
Revealing the most obscured galaxy nuclei in the mid-infrared: JWST/MIRI Spectroscopy of VV 114.
- 2023 **National Astronomy Meeting (NAM)**, Cardiff, UK, Contributed Talk
The most obscured galaxy nuclei with JWST/MIRI: the hidden growth of SMBHs?

- 2023 **The Restless Nature of AGN: 10 Years Later**, Naples, Italy, Contributed Poster
Reverberation mapping of PG 1119+120: Testing Super-Eddington Accretion.
- 2023 **National Observatory of Athens, AHEAD2020**, Athens, Greece (virtual), Invited Talk
The most obscured Galactic Nuclei in the mid-IR.
- 2022 **RAS Meeting: JWST commissioning and first science from a UK perspective**, London UK, Contributed Talk
The Obscured Nucleus of VV 114E with JWST/MIRI Spectroscopy.
- 2022 **Behind a curtain of dust, IV**, Sesto, Italy, Contributed Talk
Detecting Compact Obscured Nuclei in the Mid-Infrared.

Observing Proposals

- 2025 **CO-I**, *Unraveling Effects of Different Modes of AGN Feedback via PAH Features*, JWST (MIRI + NIRSpec). Cycle 4, 12.2 hours, PI: L. Zhang
- 2024 **CO-I**, *AGN and their outflows: probing fragmentation and survival of polycyclic aromatic hydrocarbons*, JWST (NIRSpec). Cycle 3, 18.7 hours, PI: I. García-Bernete
- 2024 **CO-I**, *Pinpointing a VLA- invisible, bright SMG in the JWST-IRAC Deep Field*, NOEMA, Grade B, 6 hours, PI: D. Rigopoulou
- 2023 **CO-I**, *Unveiling the AGN-host connection with PAH molecules.*, JWST (MIRI + NIRSpec). Cycle 2, 14.3 hours, PI: I. García-Bernete
- 2023 **CO-I**, *A JWST high-definition view of an extremely metal-poor interstellar medium.*, JWST (MIRI). Cycle 2, 31.2 hours, PI: A. Aloisi
- 2022 **CO-I**, *NOEMA and JWST unveil compact obscured nuclei in the Universe*, NOEMA, Grade C, 12.2 hours, PI: I. García-Bernete
- 2022 **CO-I**, *NOEMA and JWST unveil compact obscured nuclei in the Universe*, NOEMA, Grade B, 26.1 hours, PI: I. García-Bernete
- 2022 **CO-I**, *Pinpointing a Bright SMG for JWST Monitoring of Dusty Supernovae*, NOEMA, Grade C, 12.0 hours, PI: D. Rigopoulou
- 2021 **CO-I**, *ALMA and JWST unveil compact obscured nuclei in the Universe*, ALMA, Grade C, 30.9 hours, PI: I. García-Bernete

Other Positions

- 2024-2025 **Journal Club Organiser**, *Galaxies Journal Club*, Department of Physics, University of Oxford
- 2022-2024 **Lab Demonstrator**, *Part B Astrophysics Practicals*, Department of Physics, University of Oxford

Publications (44, 11 First Author, h-index: 19, 902 citations)

First Author

1. **Donnan, F. R.**, Sandstrom, K., Bolatto, A., et al., (2026) submitted
[JWST Observations of Starbursts: The motion of dust - PAH kinematics of M82 and NGC 253 with JWST/MIRI spectroscopy](#)
2. **Donnan, F. R.**, Ramos Almeida, C., González Martín, O., et al., (2026), A&A, 712, A72
[Kinematic detection of dusty outflows from active galactic nuclei: Polycyclic aromatic hydrocarbon kinematics of type 2 quasars with JWST/MIRI spectroscopy](#)
3. **Donnan, F. R.**, Sandstrom, K., Shivaeei, I., et al., (2026), ApJ, 1007, 52
[PAHSPECS: Spatially Resolved PAH Spectroscopy at cosmic noon with JWST MIRI MRS](#)
4. **Donnan, F. R.**, García-Bernete, I., Rigopoulou, D., et al., (2026), MNRAS, 550, stag1181
[GATOS XIV: The first direct kinematic evidence of dusty outflows from AGN through PAH kinematics of local Seyfert galaxies with JWST](#)
5. **Donnan, F. R.**, Rigopoulou, D., García-Bernete, I., et al., (2025), JATIS, 11, 031606
[Census of the most obscured galaxy nuclei over cosmic time to be revealed by PRIMA](#)
6. **Donnan, F. R.**, Rigopoulou, D., García-Bernete, I., (2024), MNRAS, 532, 75
[The kinematics of Polycyclic Aromatic Hydrocarbons \(PAHs\) in Galaxies revealed by Principal Component Analysis \(PCA\) tomography with JWST/NIRSpec](#)
7. **Donnan, F. R.**, García-Bernete, I., Rigopoulou, D., et al., (2024), MNRAS, 529, 1386
[Peeling back the layers of extinction of dusty galaxies in the era of JWST: modelling joint NIRSpec + MIRI spectra at rest-frame 1.5-28 \$\mu\text{m}\$](#)
8. **Donnan, F. R.**, Hernández Santisteban, J. V., Horne, K., et al., (2023), MNRAS, 523, 545
[Testing super-eddington accretion on to a supermassive black hole: reverberation mapping of PG 1119+120](#)
9. **Donnan, F. R.**, García-Bernete, I., Rigopoulou, D., et al., (2023), MNRAS, 519, 3691
[The obscured nucleus and shocked environment of VV 114E revealed by JWST/MIRI spectroscopy](#)
10. **Donnan, F. R.**, Rigopoulou, D., García-Bernete, I., et al., (2023), A&A, 669, A87
[A detailed look at the most obscured galactic nuclei in the mid-infrared](#)
11. **Donnan, F. R.**, Horne, K., Hernández Santisteban, J. V., (2021), MNRAS, 508, 5449
[Bayesian analysis of quasar light curves with a running optimal average: new time delay measurements of COSMOGRAIL gravitationally lensed quasars](#)

Second/Third Author

12. García-Bernete, I., **Donnan, F. R.**, Rigopoulou, D., et al., (in prep)
[\(GATOS N. High angular resolution observations of PAH bands in AGN with JWST NIRSpec and MIRI/MRS: An alternative PAH diagnostic diagram](#)

13. Thatte, N., Rigopoulou, D., **Donnan, F. R.**, et al., (2025), MNRAS, 545, 2047
[The PAH 3.4 micron feature as a tracer of shielding in the Orion Bar and NGC 6240](#)
14. García-Bernete, I., **Donnan, F. R.**, Rigopoulou, D., et al., (2025), A&A, 696, A135
[On unveiling Buried Nuclei with JWST: a technique for hunting the most obscured galaxy nuclei from local to high redshift](#)
15. García-Bernete, I., Rigopoulou, D., **Donnan, F. R.**, et al., (2024), A&A, 691, A162
[The Galaxy Activity, Torus, and Outflow Survey \(GATOS\). V: Unveiling PAH survival and resilience in the circumnuclear regions of AGN with JWST](#)
16. Rigopoulou, D., **Donnan, F. R.**, García-Bernete, I., et al., (2024), MNRAS, 532, 1598
[Polycyclic aromatic hydrocarbon emission in galaxies as seen with JWST](#)

Co-Author

17. Zhang, L., Packham, C., Hicks, E., et al., (2026), ApJL, 1007, L52
[GATOS: Distinct Feedback Modes in AGN Central Regions Revealed by Spatially Resolved JWST Spectroscopy](#)
18. Lofaro, C. M., Díaz-Santos, T., Shivaiei, I., et al., (2026)
[PAHSPECS: Polycyclic aromatic hydrocarbon properties at cosmic noon with JWST/MIRI MRS](#)
19. Rosario, D. J., Haidar, H., Campbell, S., et al., (2026)
[GATOS XV: A JWST/MIRI survey of extended circumnuclear dust emission in nearby Seyfert galaxies](#)
20. Veenema, O., Thatte, N., Rigopoulou, D., et al., (2026), MNRAS, 548, stag785
[Decoupling the AGN outflow and star-forming disc kinematics in the nuclear region of NGC 7582 with JWST NIRSpec and MIRI/MRS](#)
21. Riffel, R., Souza-Oliveira, G., Colina, L., et al., (2026), A&A, 710, A231
[Jet-driven shocks and turbulence in radio-loud Active Galactic Nuclei observed with JWST MIRI/MRS](#)
22. Pantoni, L., Baes, M., Decin, L., et al., (2026), A&A, 709, A237
[MICONIC: JWST/MIRI-MRS reveals heavily reprocessed polycyclic aromatic hydrocarbons in the circumnuclear disc of Centaurus A](#)
23. Haidar, H., Rosario, D., García-Bernete, I., et al., (2026), MNRAS, 546, stag069
[GATOS XI. Excess dust heating in the narrow-line regions of nearby AGN revealed with JWST/MIRI](#)
24. Pérez-González, P., Barro, G., Carniani, S., et al., (2026), ApJ, submitted
[Little Red Dots: One Photometric Tag Concealing Diverse Spectroscopic Flavors of Massive Star Formation and Black Hole Activity](#)
25. García-Bernete, I., Pereira-Santaella, M., González-Alfonso, E., et al., (2026), Nature Astronomy, 10, 420
[Abundant hydrocarbons in a buried galactic nucleus with signs of carbonaceous grain and polycyclic aromatic hydrocarbon processing](#)
26. Zhang, L., Packham, C., Hicks, E., et al., (2026), ApJ, 998, L32
[Evidence of Feedback Effects in Low-luminosity Active Galactic Nuclei Revealed by JWST Spectroscopy](#)

27. Ramos Almeida, C., Asensio Ramos, A., Westerdorp Plaza, C., et al., (2025), A&A, 706, A100
[Silicate emission in a type-2 quasar: JWST/MIRI constraints on torus geometry and radiative feedback](#)
28. Veenema, O., Thatte, N., Rigopoulou, D., et al., (2025), MNRAS, 544, 3361
[Shock-driven heating in the circumnuclear star-forming regions of NGC 7582: insights from JWST NIRSpec and MIRI/MRS spectroscopy](#)
29. Hunt, L. K., Draine, B. T., Navarro, M. G., et al., (2025), ApJ, 993, 84
[The Interstellar Medium in I Zw 18 Seen with JWST/MIRI. II. Warm Molecular Hydrogen and Warm Dust](#)
30. Riffel, R. A., Colina, L., Costa-Souza, J. H., et al., (2025), A&A, 705, A59
[Impact of AGN and nuclear star formation on the ISM turbulence of galaxies: Insights from JWST/MIRI spectroscopy](#)
31. Hunt, L. K., Aloisi, A., Navarro, M. G., et al., (2025), ApJ, 992, 48
[The Interstellar Medium in I Zw 18 Seen with JWST/MIRI. I. Highly Ionized Gas](#)
32. Prince, R., Hernández Santisteban, J. V., Horne, K., et al., (2025), MNRAS, 541, 642
[Echo mapping of the black hole accretion flow in NGC 7469](#)
33. Ramos Almeida, C., García-Bernete, I., Pereira-Santaella, M., et al., (2025), A&A, 698, A194
[JWST MIRI reveals the diversity of nuclear mid-infrared spectra of nearby type 2 quasars](#)
34. Hermosa Muñoz, L., Alonso-Herrero, A., Labiano, A., et al., (2025), A&A, 693, A321
[MICONIC: dual AGN, star formation, and ionised gas outflows in NGC 6240 seen with MIRI/JWST](#)
35. Zhang, L., García-Bernete, I., Packham, C., et al., (2024), ApJL, 975, L2
[Polycyclic Aromatic Hydrocarbon Emission in the Central Regions of Three Seyferts and the Implication for Underlying Feedback Mechanisms](#)
36. Pereira-Santaella, M., González-Alfonso, E., García-Bernete, I., et al., (2024), A&A, 689, L12
[H₃⁺ absorption and emission in local U/LIRGs with JWST/NIRSpec: Evidence for high H₂ ionization rates](#)
37. Haidar, H., Rosario, D. J., Alonso-Herrero, A., et al., (2024), MNRAS, 532, 4645
[Dust beyond the torus: Revealing the mid-infrared heart of local Seyfert ESO 428-G14 with JWST/MIRI](#)
38. Shen, Y., Grier, C. J., Horne, K., et al., (2024), ApJS, 272, 26
[The Sloan Digital Sky Survey Reverberation Mapping Project: Key Results](#)
39. García-Bernete, I., Pereira-Santaella, M., González-Alfonso, E., et al., (2024), A&A, 682, L5
[Structures Of Dust and gAs \(SODA\): Constraining the innermost dust properties of II Zw96 with JWST observations of H₂O and CO](#)
40. González-Alfonso, E., García-Bernete, I., Pereira-Santaella, M., et al., (2024), A&A, 682, A182
[JWST detection of extremely excited outflowing CO and H₂O in VV 114 E SW: A possible rapidly accreting IMBH](#)

41. García-Bernete, I., Alonso-Herrero, A., Rigopoulou, D., et al., (2024), A&A, 681, L7
[The Galaxy Activity, Torus, and Outflow Survey \(GATOS\). III. Revealing the inner icy structure in local active galactic nuclei](#)
42. Cackett, E. M., Gelbord, J., Barth, A. J., et al., (2023), ApJ, 958, 195
[AGN STORM 2. IV. Swift X-Ray and Ultraviolet/Optical Monitoring of Mrk 817](#)
43. García-Bernete, I., Rigopoulou, D., Alonso-Herrero, A., et al., (2022), A&A, 666, L5
[A high angular resolution view of the PAH emission in Seyfert galaxies using JWST/MRS data](#)
44. Varga, J., Hogerheijde, M., van Boekel, R., et al., (2021), A&A, 647, A56
[The asymmetric inner disk of the Herbig Ae star HD 163296 in the eyes of VLT/MATISSE: evidence for a vortex?](#)